ActInf Textbook Group ~ Cohort 3 ~ Meeting 21 (Chapter 9, part 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_3 | Cohort 3 ~ Meeting 21 (Chapter 9, part 2) | 51:27

Hi, everyone. It's Cohort 3, and August 1st, we're in our second discussion on Chapter 9.

So, if either of you would like, before we jump into some specific questions,

are there any general comments or any other topics that people want to address?

Otherwise, we will probably look through some answers and curate and improve some discourse.

Yeah, nothing from me.

All right.

I added some questions at the end, but we can go through the other stuff first.

Perfect. Okay. And Esmail, thank you for your question.

What does this text mean about epistemic value?

So, let me know if it's okay to modify the question to what is epistemic value?

If that's what your question was, or let me know.

Okay.

Let us...

You added some of the new questions at the bottom here, Neil?

Yeah, that's right.

All right.

It's one sort of big question, really.

Which one?

Let's do...

Well, in a non-Metanasiian, so that one, and then...

All right. Awesome.

Yeah.

All right. Let's start here.

All right.

Okay.

So, my take on the chapter is trying to fit active inference into a framework of what science is.

The new thing, I think, the big idea for me was the Metabasian inference, which I understand as trying to make...

Someone's trying to make inferences about something that has itself got beliefs.

So, maybe I don't know if I've got that right.

So, the first part of that question is, what does this half chapter have to contribute when that isn't the case?

If we're just trying to use it to understand something that can be modelled?

And then, the second part is what the chapter talks about, is thinking about it in the context of brains, so psychiatry.

But it presumably can be applied to anything in which the system that's being modelled can in any way be said to be of having beliefs.

And I've mentioned Dennett's intentional stance there, where we're not necessarily believing, you know, we don't think there's any sentience there.

So, you could think of...

We could talk about a thermostat thinking it's too hot, so without actually saying anything about any sort of mental processes going on there.

So, it seems to me that the Meta-Bayesian aspect could potentially be useful for this, beyond psychiatry, psychology.

Yeah, and so, skipping over the third question, which was the question is, is this actually being applied outside of neuroscience and psychology?

It seems particularly...

Well, I always thought it would be useful to be including it in behavioural economics, and maybe in systems engineering.

I don't know.

Great, great questions.

We'll start with the first one.

Oh, there we go.

Oh, okay.

I see.

The second one was here.

Also welcome, Molly.

Okay.

Thanks.

And I put the link, but this upload of the textbook,

that's a little...

It's a good starting point.

All right.

So, the first thing that...

Well...

Let's just start by recalling the Meta-Bayesian approach in 9.1.

One.

So, I think you summarized it perfectly, Neil.

We're doing Bayesian inference about something that has beliefs.

So, the aboutness of the subjective model is the environment, which may be the laboratory or the niche.

And thus, the aboutness of the scientific model is the rat in the maze, which has a model that has an aboutness of the maze.

So, that's this kind of like nested or Meta-Bayesian approach.

We're using Bayesian approaches from the outside, but then when we get to the good stuff, it is also a Bayesian model with an aboutness of an environment.

So, I think this is a great way to put it.

Trying to make inferences about something that has beliefs or trying to make inference about something that does inference.

So, your first question was, what does the chapter contribute when that is not the case?

Here, I believe we can return to our favorite path integrals paper.

And point to the continuum where on the sophisticated end of the spectrum, we have a situation like that in this figure.

Where, you know, here's us as scientists.

Here at the blanket states are the solid line.

And then we're inferring a cognitive model through, for example, structure learning or some other a priori constraints.

We're inferring the cognitive model that's looking back onto the outside.

So, the Meta-Bayesian stance is required in the cognitive on cognitive setting.

Just to be clear, to model another cognitive entity outside of the Meta-Bayesian stance, I'm not exactly sure what the alternative approach would be.

But you asked, what do we get when this is not the case?

You could model an inert system accordingly here.

For example, maybe here, you know, it's a light switch or something.

And it just, it doesn't have any action selection of its own or it's like inner or it's a passive inference object.

So, what do we gain?

We gain continuity with those cases.

Thus, grounding cognitive sciences.

We can discuss this more, but that to me feels like the big home run is continuity along this continuum of sophistication from absolutely inert particles on through arbitrarily complex cognitive entities.

So, it's not to say that it's going to be uniquely explanatory or of predictive value.

It may be like getting out an extra apparatus that you don't necessarily need.

Which would be the same thing as saying, you could have just started with a simpler model because you're bringing out this big apparatus that's just condensing down to a simpler model.

So, do any of you think of any other advantages that are gained?

So, where is that figure from?

That figure two, that paper?

Right.

Okay.

Okay.

So, it's creating, it's creating a scaffolding for the observational encounter.

That allows us to have system of interest.

Along this continuity.

It's absolutely required here.

You might be able to get away with a simpler perspective in a simpler setting.

But this is the most expressive way that we can dial it up arbitrarily complicated and also dial it back down.

Like maybe this just is a single node in this computational graph.

It just observations get passed in.

There's a single node and then it gets passed out.

But this frame, but the scientist's lab doesn't have to change.

Whether they're studying the metronome or the bird.

And I agree that it's important to connect it to these, like to the standard stages of the scientific process.

Ali or anyone else want to just add more?

If I may add some other comments about it.

But of course, I didn't hear the question from the beginning.

But from what I heard, you see, I believe it's essential to always clarify our concepts and definitions

in terms of what we're really talking about when we talk about agency on one hand.

I don't know, cognition or consciousness or whatever.

Because all of these terms, they have lots of baggage coming along with them.

So we need to be careful what exactly we mean by agency or in this case, I want to emphasize more on agency

because this is quite a controversial term.

Because as we know, at least in the mainstream biology, people don't want to bestow agency on inert objects or even some simple, quote unquote, simple organism.

But on the other hand, there are some mechanical and naturalistic definitions of agency.

Welsh's theory or from FEP and so on.

I don't know if you've seen this nice article by Philip Ball called from a couple of months ago,

Organisms as Agents of Evolution.

In this pretty much non-technical essay, he overviews some of the conceptions about agencies.

And he also, I think, yeah, he has mentioned FEP quite an extensive detail, I mean, comparatively.

But the case here, I mean, Ball's case is that we still don't know how to model agency as naturalistically as, I don't know, some physical phenomenon.

So, of course, there are possibilities to model agencies as purely physical trajectories or

probably to think about and see.

But the point here I want to make is that this is not exactly what we always think when we try to trace the path of a physical system through a state space.

We don't usually think of, I don't know, a simple particle or an inert object as following a path of or following a goal-directed motion, right?

But on the other hand, there was this recent interesting interview in Sean Carroll's Mindscape podcast with David Krakauer.

And it was quite an interesting podcast because for the first time, I heard a definition of complex systems as something that can be directly applied to FEP.

Namely, he defined complex systems or even the complexity science as the study of teleonomic matter, right?

So, obviously, what he meant by teleonomic matter doesn't have anything to do with
I don't know,
some extra material or
the
and or
these
from the
from we need
The essential...
Sorry, Ali, it just...
Sorry, just...
Okay, can you hear me now?
Yeah, just back up to teleonomic systems.
It's not an extra material cause.
It's what?
Okay.
So, yeah, it allows for the definition of agency
throughout a whole spectrum
that encompasses both inert objects or simple organisms
up to the most complex phenomenon we know
in the whole universe, namely the human brain.
So, this defining characteristic of complex systems
as teleonomic matter, I believe,
is quite an essential move here
because this can be seen
as some of the underlying assertions of FEP as well
because the same mathematical technology...
I mean, if we take Bayesian mechanics
as something that can be applied
both to a simple physical phenomenon
and to, I don't know, consciousness
or cognition or agency or whatever,
then we need to define some essential parameter
that can be put onto a continuous spectrum
without, on the one hand, limiting too much
and, on the other, without...
I mean, without including or limiting too much.
So, yeah, these recent moves

in both complexity science
and some of the recent advances in biology
seems interestingly convergent
with some of the assertions of FEP
and I hope in the future
some interesting unifying perspectives
can be gleaned from
seeing these kinds of interconnectedness
and interrelationship
between all of these theories
and specifically by looking at
how we define agency
in the most naturalistically
and mechanistic way possible.
Nice. Great.
It's like saying, like,
we can't have a framework
that has a continuity of ice
and liquid and gaseous water.
It's like, well,
it's not saying that there's not phase differences.
But we want to have something
that, like you said,
doesn't over-include or under-include
and doesn't appeal to extra material
or extra systemic factors
except credibly.
And then if we create that measure,
then we can talk about the agency
of different systems.
Okay.
So I hope...
But, you know, in your question,
you wrote,
other than unifying ideas
into a common framework.
So maybe one response is
little other than that,

but that's the big deal.
Yeah.
I mean, it's a big deal for me.
Yeah.
It's a beautiful...
It fits in, you know,
a nice bigger picture.
And I feel as though there's,
you know,
a lot of things
that could come out of that framework
thinking about it.
But it's not apparent to me
what that is
at the moment.
Yeah.
I mean, there's...
Even just the concept
that some people may walk around
feeling and projecting
that there's a difference between...
A difference in kind,
incommensurability difference
between, like,
systems that compute or don't
or cognitive systems or not,
agentic systems or not,
adaptive systems or not,
sentient systems or not.
And it's like, well,
again, we can have ice and liquid.
So it's not like we're neutering
our capacity to distinguish systems.
Literally the opposite.
Proponents...
I'm not even saying anyone specifically is,
but just allegorically.
Proponents of incommensurability

are actually stymieing
their own capacity
to compare scientifically.
They may be able to perform
qualitative comparison,
which is totally legitimate.
It is what it is, though.
But you cannot have
quantitative comparison.
Now, it may not be
as simple, simple, simple
as, like, a scalar number.
And I think that's kind of
the open space
with category theory
and different kinds of equality
and sameness.
But we need to at least
have the articulations
and the framework
to scaffold
these kinds of systems.
Otherwise,
we're just begging
for there to be
a different framework.
I mean,
if this was the periodic table,
would we want
a different framework
for different particles?
I'm sure we can think
of other ways
of where this brings
a lot of unity.
Okay.
Next question
that you added.

Metabasean inference
maybe can be applied
to understanding
any system of interest
in which the subject
can be considered
having beliefs
and intentions.
Dan,
it's intentional stance.
Yeah.
Including
into the,
even,
you have a thermostat,
but not just
a thermostat here.
It's a thermostat coupled
to an actuator unit
because you said
it can act.
But you could say
that,
yeah,
a heating system.
But this
intentional stance
goes even below
where most
of the intentional stance
people would probably
feel comfortable
and say like,
that rock
just wants to be there.
Yeah.
And so it's like,
but that's not intention.

It's like,
that's what I'm saying.
Yeah.
But either you're going to grant
there to be a continuum
of intentionality
and then I think
then whether that's
realism,
is there really
a continuum
of intentionality?
Take it or leave it.
But can we
approach systems
with a continuum
of intentionality?
Absolutely.
And that
at least topologically
has to have a structure
like this.
You could have
a different
model architecture
inside of here.
But that model
architecture
doesn't even have to be
what we would consider
necessarily a cognitive model.
Why couldn't this be
biological variables
and here's the health record
and then here's the outcomes
of health?
So this is
even more general

than cognitive modeling
per se
because rather
it's just describing
the informational
relationship
between the
modeler
and the
models
up to
and including
systems
where the
modeled
has an
aboutness
of the
laboratory
or of the
communication
itself.
I was
interested
in what
I was saying
what's the
relationship
between agency
and belief
because you could
have that
subjective model
that has
that could be said
to have beliefs
that it
might not have

agency
it might just be
medical records
for example
but those medical records
could be said
to embody
a belief
that
the
patient
has
whatever
diagnosis
Yeah
I am sure
both agency
and belief
have copious
literature
and semantics
just
off the
cuff
belief
having more
to do
with what we
would associate
with the
perceptual
sense making
side of
active inference
and agency
having to
do
more

with the
policy
selection
aspects
like
and
policy
selection
entailing
sentient
or adaptive
policy
selection
entailing
some
belief
architecture
you could
have something
that exhibits
a lot of
agency
and has
minimal
beliefs
it's just
flailing
or babbling
or thrashing
potentially
so it's
exerting
efficacy
which under
some
definitions
of agency
may qualify

with limited
belief
potentially
no justified
true belief
whereas
in their
kind of
highest
marriage
we're talking
about like
beliefs
about action
planning
as inference
those kinds
of areas
agency
it applies
action
on the
environment
I think
and as you
say
policies
it's
it's
it's
performing
some action
in a
way
that can
then help
it
learn

as opposed
to
you can
have
something
that
just
forms
beliefs
and that's
just
something
doing
passive
inference
is active
model-based
analysis
actually
being
applied
in areas
outside
of
neuropsychopathy
psychopathology
behavioral
economic
systems
engineering
so definitely
we can
bring in
some
some
papers
by
Henriksen

on
cognitive
economics
now not
all
cognitive
economics
is
utilizing
active
inference
but
some
are
this
name
it
sounds
familiar
and I
believe
it may
be
using
active
inference
little
little
but
some
industrial
systems
engineering
yes
well
robotics
another
area

industrial
systems
engineering
ranging
from the
most
logistical
and
operational
international
like
what
versus
and others
are involved
in
on through
like what
Stephen Fox
is involved
in
with
social
organization
so here's
this
then here
we have
the
Stephen Fox
John Boyk
angle
with
socio-technical
systems
um
nagro
ecology

in fact
I
I'm gonna
go to
another
this is a
little
little peep
into our
uh
into our
open source
ecosystems
document
um
which of
course
Ali and
Neil you're
you're welcome
to be authors
on just
just email
me if you
if you don't
have the
information
that you feel
like you
need
um
this is our
citation train
I think we
went from
we go from
yeah literally
citation 7

to 99
so it's
it's kind
of a monster
cognitive
neuroscience
and cognitive
philosophy
artificial
intelligence
and AI
explainability
robotics
human health
clinical
psychiatry
social
developmental
psychology
social
sciences
mathematical
physics
physics
of life
consciousness
studies
phenomenology
evolutionary
ecology
and development
cyber
and cognitive
security
mapping
and modeling
of information
ecosystems

ontology
modeling
and maintenance
rhetorical
analysis
logistics
and business
intelligence
theoretical
and applied
biology
and ecology
project
management
and team
science
collective
behavior
military
science
if there
are some
other domains
we will
literally
add them
in fact
we probably
should
add
economics
I know
that we
cite some
though
but also
I'll just
write

social
sciences
and
economics
any
thoughts
on this
list
or
like what
other
domains
would be
exciting
for you
to see
that's
that's
yeah
that's
really good
I'm just
gonna
I'm just
gonna copy
it into
the chat
too
yeah
that's too
long
it's too
long
I mean
but
just
I'm
gonna

copy
the
document
there
but
yeah
of course
we're in
textbook groups
I'm not gonna
go down the
rabbit hole
with the
open source
ecosystems
but
that's
exactly
what we've
been thinking
about
and also
we didn't
even like
cite
digital
Gaia
verses
and so
on
or
Thought
Forge
or any
of our
other
colleagues
in

industry
and that's
kind of
the idea
with the
ecosystem
we have
people who
are working
more on
the theory
and then
it's not
always so
visible
necessarily
where the
applications
are happening
but I
think we
can do
a lot
of work
to improve
like the
discoverability
and connect
you know
use the
ontology
to connect
across
settings
and cases
there's
there's
just a

lot
but
again
another
conversation
all right
all right
Esmail
asked
what is
epistemic
value
let's
see where
it comes
up in
chapter
nine
otherwise
we can
look to
an earlier
chapter
as well
yeah
epistemic
values
primarily
discussed
in
in
chapter
seven
Ali
what's
your take
on
epistemic

value
and Neil
as well
let's
just get
some
looking
from chapter
nine
looking back
to seven
what is
epistemic
value
okay
so
epistemic
value
is
one of
the
key
terminology
of active
inference
inference
which is
also
in
parallel
with
I mean
practical
value
because
you see
sometimes
when an

agent
tries to
infer
its best
course of
action
within an
environment
it's
not always
a trade
off
between
I mean
the
thing
that
or
it's
it's
not
a
necessary
trade
off
between
survival
and
also
the
explore
exploit
phenomena
so
for example
in
some
of the

cases
the
agent
needs
to
somehow
gather
information
about
the
environment
and it
actually
prefers
gathering
the
information
about
the
environment
and
the
situation
in which
it resides
in order
to
take
more
suitable
actions
for it
to
preserve
its
persistence
through
time

or in
other words
its
survival
so
epistemic
value
is
what
gives
a kind
of
numerical
value
to
this
kind
of
exploring
the
environment
and
also
how
we
can
use
this
numerical
value
in a
trade-off
between
what we
gain
informationally
about the
environment

and the
course of
actions
we take
through
within that
environment
so
sometimes
it
acts as
a central
essential
goal or
purpose
for the
agent
to
increase
its
epistemic
value
through
its
information
seeking
actions
rather than
just
changing
the
environment
in order
to
better
fit
its
generative

model
nice
thank you
Neil
what's
your
take
oh
you're
muted
if
you're
adding
anything
or
all
sorry
I was
yeah
I was
thinking
when
I was
talking
I
think
of
the
things
like
exploration
exploitation
so in
reinforcement
learning
exploration
it's
it's

very
much
exploration
is
for
direct
impact
on the
policy
but we
as humans
go through
life
collecting
information
that
and storing
it
with
likely
no future
use of
it
so you
could say
we're
epistemic
we do
epistemic
gathering
of
stuff
that
has
no
apparent
epistemic
value

but
circumstances
might
change
and it
might
become
valuable
so I
don't know
if there's
a distinction
to be
made
between
well
yeah
if
bringing
in this
idea
of
value
of
knowledge
yeah
that
that's
a very
interesting
point
about
the pure
value
of
information
at least
with respect

to its
role
in decision
making
like if
we're
thinking
about
people
using
social
media
applications
and so
on
or
who knows
what
else
maybe
even
just
the
desire
to
flip
every
rock
over
see
if
there's
an
ant
colony
underneath
there
like

there
are
expected
free
energy
being
composed
of
pragmatic
and
epistemic
value
gives
us
extreme
flexibility
to
model
a
diversity
of
settings
and
cases
so
this
is
summarized
in
figure
2.6
on the
left
of the
main
equation
we have
pragmatic

value
which is
how
surprising
observations
are
with respect
to
preferences
and on
the right
side
we have
the
larger
epistemic
value
which is
a
KL
divergence
hence
can be
tractably
optimized
in the
variational
Bayesian
framework
and
3
and
5
are
identical
whereas
2
and

4
differ
and
specifically
4
brings
in
the
observations
as
conditioned
upon
that
specific
policy
being
enumerated
or calculated
for expected
free
energy
so
under
a
policy
where
the
observations
are
not
expected
to
add
any
value
in
terms
of

reducing
uncertainty
that
policy
is
not
assigned
a
high
epistemic
value
it may
have
a
high
pragmatic
value
or not
whereas
a
sequence
of
observations
that
a
given
policy
provides
which
do
cause
a
divergence
from
what
is
understood
other

than
through
that
action
being
taken
that
would
be
informative
again
that
may
have
a
positive
or
detrimental
pragmatic
value
like
what's
I wonder
what would
happen
if I
did
this
but
that
may
threaten
homeostasis
but
having
epistemic
and
pragmatic

value
unified
under
a
singular
decision
making
imperative
allows
us to
find
these
previously
unappreciated
concepts
in
in
a
to
be
under
the
umbrella
of
expected
free
energy
which
we
utilize
for
policy
selection
but
most
briefly
epistemic
value

is the
value
of
incoming
information
in
terms
of
observations
about
hidden
states
reducing
uncertainty
about
the
hidden
states
chapter
seven
deals
with
it
more
one
interesting
question
maybe
for the
future
is like
how is
epistemic
!
value
treated
differently
in the

discrete
and in
the
continuous
case
like
do
you
need
explicit
planning
discrete
potentially
maybe
even
symbolized
planning
in
order
to
truly
engage
with
the
counterfactual
value
of
information
so
how do
we
think
about
the
epistemic
value
of
continuous

activity
like
groping
I mean
there might
be a
connection
there to
optimal
grasp
and
groping
around
in a
sort of
continuous
reflexive
way
to gain
a grip
on an
object
but
with
that
load
on
pragmatic
value
where
optimal
grip
is
preferred
and
or
with
that

load
on
epistemic
value
where
how
to
achieve
optimal
grasp
or
what
it
would
feel
like
to
have
optimal
grasp
is
not
known
what
are
the
two
related
reasons
for
fitting
a
computational
model
to
observed
behavior
believe

this
is
an
exact
quote
yeah
let's
uh
okay
let's
see how
well
perplexity
dot
ai
does
it's
gonna
have
to
context
switch
a
little
bit
it's
gonna
have
to
change
languages
but
we
all
do
i
think
this

is
also
which
more
and
more
synthetic
intelligences
are
gonna
now
it's
asking
us
a
question
now
how
would
it
evaluate
all the
questions
that it
could
ask
us
which
one
should
ask
well
active
inference
how
will
it
know

what
question
to
ask
what
path
of
action
is
gonna
be
most
relevant
okay
here
we
go
here
we
go
first
estimate
parameters
of
interest
in
terms
of
computational
phenotyping
the
second
is
to
compare
alternative
hypotheses
so

like
to
give
an
example
from
from
botany
um
the
first
one
would
be
like
we
have
a
bunch
of
strains
of
corn
and
we
want
to
know
which
one
has
a
different
growth
rate
computational
phenotyping
the

second
one
would
be
like
does
corn
growth
rate
depend
on
just
humidity
or
humidity
and
genotype
or
humidity
genotype
and
temperature
so
the
first
one
is
parameter
estimation
computational
phenotyping
is
parameter
estimation
and
the
second
reason

the
comparison
of
alternative
hypotheses
is
model
comparison
the
synthetic
intelligence
here
has
provided
a
different
set
of
answers
that
aren't
terrible
and
it
can't
necessarily
answer
perfectly
from
the
textbook
anyway
9.1
wow
Ollie
this
looks
like

somewhat
of a
little
side
journey
I
don't
remember
it
but it
seems
like
it
happens
foundational
ists
coherent
infinitists
that's
good to
have
about
that
this
goes
beyond
proof
of
principle
simulations
we have
seen
in
previous
chapters
instead
exploits
active

inference
in
answering
scientific
questions
what does
that
mean
this
is
a
pretty
concise
answer
previous
chapters
generated
the
model
and
played
around
with
it
e.g.
used
it
to
generate
plausible
data
now
we are
building
the
machinery
to
take

in
real
data
and
fit
that
to
data
and
fit
that
data
to
a
model
to
slash
with
a
model
to
uses
on
page
174
map
territory
fallacy
okay
not
a
full
question
how
does
this
work
not

how
does
the
work
connect
with
map
territory
fallacy
in
the
context
of
chapter
nine
being
about
data
analysis
does
the
work
commit
the
map
territory
fallacy
or
rather
help
us
articulate
or evade
being
fallacious
so
let's
think

about
figure
9.1
some
have
criticized
FEP
by
implying
or
suggesting
it's
guilty
of
making
a
fallacy
that
the
map
is
the
territory
I
mean
citations
needed
also
regarding
generative
process
and
generative
model
we
actually
spoke
with

Maxwell
in
the
scientific
advisory
board
meeting
it was
really
funny
and
of
course
illuminating
of high
epistemic
value
here's
what he
added
this
morning
I
of
course
pointed
to
Ali's
question
during
the
inner
screen
stream
and
he
said
there's

one
generative
model
it's
the
system
level
description
it's
our
model
of
the
whole
environment
agent
system
it is
the
entire
causal
connective
tissue
it's
a
joint
density
of
the
real
model
of
the
dependencies
of
the
system
if

the
generative
model
contains
a
markup
blanket
via
sparsity
and it
does
then you
get an
additional
density
the
variational
density
where
internal
states
parameterize
with an
aboutness
of
external
states
and those
internal
states
will look
like
they
are
tracking
external
things
in the

sense
of the
generalized
synchrony
generative
process
refers to
the
dynamics
that
give
rise
to
observations
that
are
hard
coded
in
and so
he
analogized
that
to
the
Newton's
laws
it is
not
the
case
that
the
generative
model
is a
model
of the

generative
process
philosophers
are comfortable
with approaches
like the
Helmholtz and
the
Boltzmann
machine
and from
there
arose
the
distinction
between
the
recognition
and the
generative
density
table
two
densities
but that
is not
the sense
in which
generative
model is
being used
here
so just
kind of a
fun
update
to an
ongoing

discussion
and
and
Carl
was
there
and
endorsed
and
approved
well
and
time
flies
time
flies
like
an
arrow
fruit
flies
like
a
banana
as
the
famous
quote
said
we are
at the
end of
chapter
nine
now
in
the
coming

weeks
we will
be
heading
into
chapter
10
then
after
two
weeks
on
the
text
and
concept
heavy
chapter
10
we will
be
closing
this
cohort
so
it's
very
exciting
presumably
after
this
we
will
wrap
the
transcripts
of
cohort

one
two
and
three
and
publish
those
three
together
anyone
have
any
closing
thoughts
good
session
and
I
suspect
next
time
chapter
10
looks
good
awesome
yep
totally
looking
forward
to
it
all
right
thank
you
fellas
till

next

time

thanks

cheers

bye

bye